# Ayush Gaggar

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## Education

**Northwestern University** **2022 - 2027**
*Ph.D. Candidate in Mechanical Engineering (Robotics)* *Chicago, IL*

**University of California, San Diego** **2020 - 2022**
*B.S. Mechanical Engineering, Minor in Entrepreneurship & Innovation, GPA: 3.96/4.00* *La Jolla, CA*

## Research & Industry Experience

**Ph.D. Research Intern, Multimodal AI** **May 2026 - Aug. 2026**
*Dolby Laboratories, Inc.* *San Francisco, CA*

- Adapted Meta's SAM-Audio foundation model to detect off-screen sound sources in egocentric video, a key step toward closed-loop perception where an agent turns toward what it hears but cannot see.
- Designed InterMaskAttention, a cross-attention layer that enables joint decomposition into per-object stems and a single off-screen residual, improving zero-shot off-screen detection precision by ∼65% over LoRA fine-tuning.
- Built the full pipeline end-to-end, including SAM2 auto-masking, a synthetic off-screen dataset from AVSBench, and LoRA fine-tuning, doubling separation quality (CLAP) over frozen SAM-Audio.

**Ph.D. Research Intern, Cooperative Cognition Science** **Feb. 2026 - May 2026**
*Student Associate at Honda Research Institute, Inc. (HRI)* *San Jose, CA*

- Developed a framework for reasoning over the joint distribution of multiple agents' futures, coupling each agent's trajectory distribution via unbalanced optimal transport (OT) within closed-loop model predictive control (MPC).
- Demonstrated on two-agent PushT co-manipulation that OT coordination recovers from local minima, including emergent pivoting behavior, with ∼20% higher success and ∼15% faster completion than independent control.
- Researched a formal measure of when coordination is required, independent of environment encoding task success.

**Graduate Researcher, Northwestern University** **Jan. 2023 - Present**
*Interactive & Emergent Autonomy Lab — PI: Prof. Todd Murphey* *Evanston, IL*

- Investigating autonomous data collection to improve latent world model learning for robotic skill acquisition.
- Studying robot decision-making from early latent representations, treating predictive uncertainty as a signal for robust planning rather than a quantity to minimize.
- Previous work includes low-data NeRFs & Gaussian Splatting, linear-complexity ergodic control, dual-arm manipulation, and benchmarking on the Franka Emika arm.
- Technical focus: robot perception, world models, diffusion models, VLMs/VLAs, runtime learning, and embodied AI.

**Undergraduate Researcher, University of California, San Diego** **Jan. 2021 - Jun. 2022**
*Bioinspired Robotics Design Lab — PI: Prof. Mike Tolley* *La Jolla, CA*

- Fabricated a squid-inspired underwater robot, evaluating four differential body geometries for locomotion.
- Designed and evaluated five cam profiles using MATLAB-based power and thrust analysis, validating propulsion performance through Arduino-controlled underwater displacement and velocity experiments.

**Mechanical/Thermal Engineering Intern** **Jun. 2021 - Sep. 2022**
*Advanced Micro Devices, Inc. (AMD)* *Sunnyvale, CA*

- Designed a test fixture to verify EPYC servers socket pin connectivity under static loads up to 50 kgf.
- Developed a retimer chip cooling assembly deployed in 200+ servers, including BOM and manufacturing documentation.
- Prototyped socket handling trays that accelerated the Genoa launch while preventing damage to up to 50 server sockets.

## Publications

[1] Max Muchen Sun, **Ayush Gaggar**, Peter Trautman, and Todd Murphey. "Fast Ergodic Search with Kernel Functions." *IEEE Transactions on Robotics*, 2025.

[2] **Ayush Gaggar**, Todd Murphey. "Data Augmentation for NeRF in the Low Data Limit." *International Conference on Robotics and Automation (ICRA)*, 2025.

[3] Patent: Max Muchen Sun, **Ayush Gaggar**, Peter Trautman, and Todd Murphey. "Kernel-Based Ergodic Search." SN: 18/914929. 2024.

## Technical Skills

**Embodied AI**: PyTorch, JAX, Latent World Models, Diffusion Models, NeRFs, Gaussian Splatting, GPs
**Robotics**: ROS/ROS2, MuJoCo, Gazebo, PyBullet, Optimal Control, Franka Emika, HDT Adroit, AgileX Piper
**Engineering**: Python, C++, MATLAB, Linux, Git, SolidWorks, Fusion 360 (CAD/CAM), Simulink, LTSpice, CNC, FDM 3D Printing, Soldering, Instron

## Selected Engineering Projects

**EKF SLAM for Differential Drive Robot** **Jan. 2023 - Mar. 2023**

- Implemented Extended Kalman Filter SLAM from scratch in ROS2 C++, using LiDAR data to localize a Turtlebot3.
- Integrated odometry readings, supervised and unsupervised learning from LiDAR data, and created a simulation environment in RViZ2 for the full functioning SLAM algorithm.

**Robot Play: Don't Let the Balloon Touch the Ground** **Nov. 2022 - Dec. 2022**

- Controlled a 7DOF Franka arm to track, predict, move to, and hit a balloon such that it would never hit the ground.
- Developed a ROS2 control system from scratch, including forward and inverse kinematics in cartesian and joint space.
- Employed a RealSense RGB-D camera to isolate the centroid of moving, red objects and forward predict its motion.

**360° Scanning Sound Array** **Jan. 2022 - Jun. 2022**

- Designed a novel testing apparatus to measure noise (dB and freq.) emitted by small-scale UAVs for industry sponsor ATA Inc. as a function of drone's 3DOF moment in 3D space, primarily for national defense interests.
- Primarily manufactured using manual and CNC mills and lathes, with FEA analysis for a 4.24 factor of safety.
- Implemented open loop motor control to determine microphones' spherical positioning with an accuracy of $\pm 0.015°$.

## Awards

Doctoral Consortium, *International Conference on Robotics & Automation* **May 2025**
Murphy Fellowship, *Northwestern University* **Sep. 2022**
Magna Cum Laude, *University of California, San Diego* **Jun. 2022**
Best Student Presenter, *Project in a Box, UCSD Undergraduate Conference* **May 2021**
2x Dean's and President's Scholar, *San José State University* **May 2019**
Eagle Scout, *Boy Scouts of America* **Aug. 2018**

## Leadership & Services

**Teaching Assistant** **Fall 2024, Spring 2025**

*Northwestern University, Department of Mechanical Engineering* *Evanston, IL*

- Assisted in an upper-division Dynamics class with 80 students each quarter with grading and office hours.
- Guest lectured on dynamics concepts like the Lagrangian and Special Rotation matrices.
- Overwhelmingly positive student evaluation (96%) of skill in communication and engagement.

**Private Tutor & Peer Educator** **2017 - 2025**

*UCSD Dept. of Engineering & Freelance* *Multiple Locations*

- Mentored 125+ engineering students in mathematics and physics through office hours, review sessions, and independently developed instructional materials.

**Museum Presenter - Volunteer** **2023 - 2025**

*Museum of Science & Industry* *Chicago, IL*

- Presented my robotics research during national robotics week, where thousands of children from grades K-12, along with adults, interacted with the experimental systems.